Needs Assessment Report

Project Title : Development of Integrated E- Library Management System

Client Organization: Ethiopian Geological Institute (EGI)

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1. Introduction

The Ethiopian Geological Institute (EGI) manages large volumes of geological, spatial, laboratory, and administrative information generated from different operational units and research activities. These datasets are critical for mineral exploration, environmental studies, geological analysis, institutional planning, and national development initiatives.

Currently, data resources are distributed across multiple storage environments including local computers, spreadsheets, printed documents, external storage devices, and departmental repositories. This fragmented environment creates operational inefficiencies, slow data retrieval processes, limited collaboration, inconsistent metadata management, and weak accessibility for stakeholders.

This Needs Assessment Report presents the institutional and technical needs identified during preliminary assessment activities and stakeholder discussions. The report focuses on requirements related to data collection and ingestion, database development, data storage, data management, online/web-based services, analytics, and decision-support information generation.

The findings of this assessment will guide the design and implementation of a centralized and scalable geological information management system.

2. Objectives of the Needs Assessment

The main objective of the assessment is to identify institutional, operational, and technical requirements necessary for developing an integrated digital information management platform.

Specific Objectives

- Assess current challenges in data handling and information management.
- Identify requirements for data collection and ingestion.
- Determine database and storage requirements.
- Identify institutional needs related to data sharing and accessibility.
- Assess requirements for analytics and reporting services.
- Define online and web-based service expectations.
- Support future system architecture and implementation planning.

3. Assessment of Current Challenges

The assessment identified several operational and technical limitations affecting institutional performance and information management efficiency.

Fragmented Data Environment

Data resources are distributed across different departments and storage mechanisms without centralized coordination. This limits accessibility and increases duplication risks.

Manual Processes

Many data management activities are performed manually, including document handling, report retrieval, metadata entry, and file sharing.

Limited Search and Retrieval Capability

Existing systems do not provide advanced metadata indexing, GIS-based search, or full-text document retrieval services.

Weak Data Sharing Mechanisms

Departments face difficulties in securely sharing data internally and externally with researchers, mining companies, and stakeholders.

Lack of Commercial Service Integration

The institution currently lacks mechanisms for secure online sales of premium geological reports and datasets.

Limited Decision-Support Tools

Existing environments provide limited analytics, reporting, and dashboard capabilities for management and technical decision-making.

4. Client Needs Assessment

4.1 Data Collection and Ingestion Needs

The client requires an efficient and scalable mechanism for collecting and ingesting different types of geological and institutional datasets.

Identified Needs

- Support for multiple file formats including PDF, GIS, Excel, images, and multimedia.
- Bulk upload functionality for large datasets.

- Metadata capture during upload processes.
- Automated validation of uploaded records.
- GIS and spatial data ingestion support.
- Version control for uploaded documents.
- Secure upload mechanisms with user authentication.

The proposed system should simplify data ingestion while maintaining metadata quality and consistency.

4.2 Database Development Needs

The institution requires a centralized and scalable database capable of handling complex geological data relationships and institutional records.

Identified Needs

- Centralized PostgreSQL database architecture.
- Custom geological metadata schema.
- GIS spatial database support.
- Advanced indexing and filtering mechanisms.
- Integration with search engines such as Elasticsearch.
- Scalable architecture for future institutional growth.
- Support for relational and spatial datasets.

The database structure should support geological reports, GIS layers, laboratory records, multimedia files, and administrative information.

4.3 Data Storage Needs

The institution requires secure and reliable storage infrastructure capable of supporting long-term preservation and accessibility of digital resources.

Identified Needs

- Centralized digital repository.
- Secure file storage services.
- Automated backup and recovery mechanisms.
- Large file storage support.
- Cloud-ready architecture.
- Redundancy and disaster recovery planning.
- Secure archival storage for historical documents.

The proposed system should improve institutional data preservation and reduce risks associated with fragmented storage practices.

4.4 Data Management Needs

The assessment identified strong institutional demand for improved data governance and management services.

Identified Needs

- Role-based access control (RBAC).
- Metadata management and standardization.
- Data validation and approval workflows.
- Audit trails and activity monitoring.
- User management services.
- Document categorization and classification.
- Data quality monitoring mechanisms.

The institution also requires secure authentication systems and controlled access to sensitive geological information.

4.5 Online and Web-Based Service Needs

The institution requires modern web-based services that improve accessibility, collaboration, and public engagement.

Identified Needs

- Responsive web portal.
- Advanced search and filtering services.
- Full-text search capability.
- GIS visualization and spatial search.
- Public and restricted access environments.
- Online report purchasing and downloads.
- Integration with Telebirr and Chapa payment systems.
- User dashboards and activity tracking.
- API services for future interoperability.

The proposed platform should provide both public information access and secure institutional services.

4.6 Data Sharing Needs

The institution requires secure and controlled mechanisms for sharing geological information internally and externally.

Identified Needs

- Controlled sharing of datasets between departments.
- Secure public access to approved resources.
- Download management services.
- Access permissions based on user roles.
- External stakeholder access management.
- Data exchange support for research collaboration.

Improved sharing mechanisms will enhance institutional collaboration and service delivery.

4.7 Data Transformation and Analytics Needs

The institution requires tools capable of transforming raw datasets into meaningful information for operational and strategic purposes.

Identified Needs

- Dashboard and visualization tools.
- Statistical reporting services.
- GIS-based analysis.
- Metadata analytics.
- Document usage statistics.
- Data trend analysis.
- Reporting tools for management decision-making.

The proposed system should support evidence-based planning and institutional reporting activities.

5. Target Beneficiaries

The proposed system will support multiple stakeholder groups including:

- EGI researchers and technical staff
- Geological experts and survey teams
- Mining and exploration companies
- Universities and academic researchers
- Government organizations
- International consultants

- Public users accessing approved resources

The platform will improve accessibility, collaboration, and information service delivery across all stakeholder groups.

6. Proposed Solution Direction

Based on the assessment findings, a custom-built web-based information management platform is recommended.

Recommended Technology Stack

Component	Technology
Frontend	React.js
Backend	Node.js
Database	PostgreSQL
Search Engine	Elasticsearch
GIS Integration	GeoServer/QGIS
Authentication	JWT/OAuth
Payment Integration	Telebirr /CBE

Justification for Custom Development

A custom-built solution is recommended because it provides:

- Flexible geological metadata structures.
- Better GIS integration capabilities.
- Modern responsive user experience.
- Direct payment gateway integration.
- Scalability for future institutional expansion.
- Better customization for geological workflows.

7. Conclusion

The Needs Assessment confirms the strong institutional requirement for a centralized, secure, and scalable geological information management platform capable of supporting data ingestion, storage, management, sharing, analytics, and web-based service delivery.

The proposed system will significantly improve operational efficiency, accessibility, collaboration, institutional data preservation, and decision-making capabilities within the Ethiopian Geological Institute.